Xingyu Wang

Nationality: Chinese **Date of birth:** 4 Jan 2000 **Gender:** Male **Phone:**

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CAREER OBJECTIVE

Career focus

Dedicated Computer Engineer specializing in IoT and AI-driven systems, with strong programming skills in C++, Python, and Java. Seeking to leverage my expertise in software design and development to contribute to innovative technology projects in a dynamic company.

EDUCATION & TRAINING

Master of Computer Engineering

Paderborn University [1 Apr 2022 - Current]

City: Paderborn | **Country:** Germany

Website: https://www.uni-paderborn.de | **Field(s) of study:** IoT Systems, Embedded Systems, and Machine Learning:

Bachelor of Computer Engineering

Assumption University, Faculty of Engineering [30 Sep 2017 - 1 Oct 2021]

City: Samutprakarn | **Country:** Thailand

Website: https://www.au.edu | **Field(s) of study:** Computer Engineering: | **Final grade:** 3.72/4.00

SKILLS SECTION

Technical Skills

- **Programming & Development**:

Python (Advanced), C/C++ (Embedded), OpenCV

- **IoT & Embedded Systems**:

Raspberry Pi, Riotee Boards, RIOT OS, nRF52840, BLE Communication Protocol

- **Electrical Engineering & Automaton:**

Sensor Integration, Circuit Design, Battery-Free Systems, IT Infrastructure, Edge Computing Architecture

- **AI & LLMs:**

LLM fundamentals & underlying mechanisms, Agent-based workflows & reusable skills, Prompt Engineering

PROJECTS

[15 Jan 2025 - 15 Jan 2026]

Work Experience - Digital Village Twin Etteln (DiDoZ Etteln) Machine Learning & IoT Project | [Software Innovation Campus Paderborn]

Project Description: architected and implemented a real-time data pipeline for smart city environments, processing time-series sensor data from different sensors, dynamic operational mode switching (low-latency vs. low-data-volume modes) based on anomalies detection.

As a student research assistant:

- Implemented edge computing architecture to process high-velocity sensor time-series data streams.
- Engineered pattern detection algorithms (DeepAnT - A Deep Learning Approach for Unsupervised Anomaly Detection in Time Series) to dynamically adjust operational modes based on data urgency and identified anomalies.

Technologies: Python, PyTorch, DeepAnT, Streamlit, Real-Time Stream Processing, Edge Computing

[1 Oct 2024 - 1 Oct 2025]

Project Group - Multi-Node Environmental Monitoring Demonstrator IoT Systems Project | [Paderborn University]

Project Description: designed and deployed a battery-free IoT network with 4 Riotee sensor nodes, one coordinator, and e-paper display for real-time environmental monitoring

As a member of three-person team:

- Performed electrical integration including power calculations for energy harvesting systems, capacitor sizing, and cabling for multi-node physical setups
- Developed embedded firmware in C/C++ for ultra-low-power microcontrollers (nRF52840), managing sensor interfacing (SHTC3 temperature/humidity sensors), displaying temperature, humidity, and network status with dynamic updates

Technologies: BLE Protocol, C/C++, Python, Energy Harvesting, Embedded Systems

LANGUAGE SKILLS

Mother tongue(s): Chinese

English

LISTENING: C1 **READING:** C1 **WRITING:** B1

SPOKEN PRODUCTION: B1

SPOKEN INTERACTION: B2